Northstar Logistics

Warehouse Management & Logistics Performance Analysis

Project: SQL portfolio project
Database: SQLite
Analysis period: January–June 2026
Tools: SQL, DB Browser for SQLite

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1. Project overview

Northstar Logistics is a fictional logistics business operating three distribution centres:

- WH01 – Corby Distribution Centre
- WH02 – Leicester Fulfilment Hub
- WH03 – Birmingham Regional Hub

The purpose of this project was to use SQL to investigate operational data and identify potential issues affecting inventory accuracy, order fulfilment and delivery performance.

My analysis focused on five key areas:

1. Inventory accuracy and stock continuity.
2. Order cancellations and customer patterns.
3. Warehouse operational performance.
4. Carrier and delivery performance.
5. Practical recommendations for improving logistics operations.

The aim was not simply to produce SQL queries, but to use the results to identify business problems and suggest areas for further investigation.

2. Executive summary

The analysis identified several areas that Northstar Logistics should investigate further.

Inventory continuity: Daily stock calculations were internally consistent across 16,290 records. However, an investigation of SKU013 at Corby identified unexplained increases in opening stock where no receipts were recorded.

Order cancellations: The overall cancellation rate was approximately 3.63%. Birmingham had the highest warehouse cancellation rate at 4.86%, while February had the highest monthly rate at 5.38%.

Warehouse delivery performance: Leicester had the highest late-delivery rate at 28.26%, compared with 20.88% at Corby and 19.78% at Birmingham.

Carrier performance: National Freight had the highest late-delivery rate across all three warehouses. Its late-delivery rates ranged from 33.71% in Birmingham to 42.53% in Leicester.

Customer cancellations: Some customer and warehouse combinations had noticeably higher cancellation rates. However, these were often based on relatively small order volumes.

These findings provide several opportunities for further investigation, particularly around inventory movement records, Leicester's dispatch processes and National Freight's delivery performance.

3. Analysis and findings

3.1 Inventory accuracy and stock continuity

Inventory balance validation

Investigation: Tested whether the daily inventory calculations reconciled using opening stock, units received and units used.

SQL approach: Compared the recorded closing stock against the calculated balance:

```
Opening stock + Units received - Units used
```

Finding:

- Total inventory records: 16,290.
- Records matching the expected calculation: 16,290.
- Mismatched records: 0.

This confirms that the recorded closing stock matched the expected calculation across the dataset.

Inventory data-quality checks

Checks included missing stock values, zero-stock records, negative quantities and duplicate daily inventory combinations.

Four records had zero closing stock. No negative stock values, missing key fields or duplicate combinations were identified in the checks performed.

These checks provide confidence in the basic structure of the inventory data, although they do not prove that every stock movement has been recorded.

Inventory continuity investigation – SKU013, WH01

Investigation: Compared each day's opening stock against the previous day's closing stock using the SQL window function `LAG()`.

Finding:

Between 17 and 20 March 2026, opening stock increased from 11 to 21 units, despite the previous day's closing stock being zero. No stock receipts were recorded on these dates.

The daily stock calculations were internally consistent, but the opening balances did not reconcile with the previous day's closing balances.

Why it matters:

Unrecorded stock movements or adjustments could affect inventory accuracy and stock availability.

Recommended follow-up:

Investigate whether stock adjustments, transfers or other movements were omitted from the available records. The cause cannot be confirmed from the current dataset.

3.2 Order cancellations and customer patterns

Overall cancellation performance

Investigation: Compared order volumes and cancellation rates across warehouses and months.

The dataset contained 1,434 orders:

- Shipped: 1,382.
- Cancelled: 52.
- Overall cancellation rate: approximately 3.63%.

Cancellation rate by warehouse:

Warehouse	Orders	Cancelled	Cancellation rate
Corby (WH01)	491	12	2.44%

Leicester (WH02)	470	17	3.62%
Birmingham (WH03)	473	23	4.86%

Finding:

Birmingham had the highest cancellation rate, almost twice Corby's rate. Corby processed the most orders and had the lowest cancellation rate.

The reasons behind the differences are not established by the available data.

Monthly cancellation trends

February had the highest cancellation rate:

- February: 12 cancellations from 223 orders – 5.38%.
- January: 6 cancellations from 257 orders – 2.33%.
- May: 8 cancellations from 252 orders – 3.17%.

The cancellation rate declined after February before increasing again in June to 3.98%.

Recommended follow-up:

Investigate whether February's cancellations were concentrated among particular customers, warehouses or order channels.

February cancellations by order channel

Channel	Orders	Cancelled	Cancellation rate
EDI	48	3	6.25%
Email	69	4	5.80%
Phone	53	3	5.66%
Portal	53	2	3.77%

EDI had the highest cancellation rate, while Email had the highest number of cancellations.

EDI cancellation rates also varied throughout the first half of 2026, peaking in February at 6.25%.

Finding:

There is no clear evidence that one order channel is responsible for cancellations. The results suggest that channel performance should be monitored over time before any operational changes are recommended.

Customer cancellation patterns

Further analysis compared cancellation rates by customer, warehouse and order channel.

Some customer and warehouse combinations showed higher cancellation rates:

- Business Customer 34 at Leicester: 3 cancellations from 13 orders – 23.08%.
- Business Customer 24 at Birmingham: 2 from 9 – 22.22%.
- Business Customer 13 at Leicester: 2 from 12 – 16.67%.

Business Customer 42 also had cancellations across multiple warehouses, including Corby and Birmingham.

The additional order-channel analysis did not identify one consistent channel associated with these customers' cancellations.

Finding:

Cancellations appear concentrated in some customer and warehouse combinations, but the relatively small order volumes mean that the percentages should be treated cautiously.

Recommended follow-up:

Review cancellation reasons and customer order histories to establish whether there are recurring issues, such as order-entry errors, stock availability or customer-requested cancellations.

3.3 Warehouse delivery performance

Investigation: Compared late deliveries across the three warehouses using actual delivery dates and promised delivery dates.

Warehouse	Shipments	Late deliveries	Late rate
Corby (WH01)	479	100	20.88%
Leicester (WH02)	453	128	28.26%
Birmingham (WH03)	450	89	19.78%

Finding:

Leicester had the highest late-delivery rate at 28.26%, noticeably higher than the other two warehouses.

Birmingham had the lowest late-delivery rate at 19.78%.

An interesting contrast is that Birmingham had the highest cancellation rate but the lowest late-delivery rate. This suggests that cancellations and late deliveries may involve different operational issues.

Recommended follow-up:

Investigate Leicester's dispatch processes, carrier handover times and delivery-area patterns to understand why its late-delivery rate is higher.

3.4 Carrier delivery performance

Investigation: Compared delivery times, on-time performance and late-delivery rates across the four carriers.

Average delivery time

Carrier	Shipments	Average delivery time
National Freight	288	2.42 days
ParcelNorth	362	2.29 days
RapidRoute	336	2.16 days
SwiftShip	396	2.08 days

National Freight had the longest average delivery time, while SwiftShip had the shortest.

On-time delivery performance

Carrier	Shipments	On time	On-time rate
National Freight	288	177	61.46%
ParcelNorth	362	269	74.31%
RapidRoute	336	287	85.42%
SwiftShip	396	332	83.84%

RapidRoute recorded the highest on-time delivery rate, despite SwiftShip having the shortest average delivery time overall.

This demonstrates why average delivery time and on-time delivery percentage should be considered separately.

On-time versus late delivery times

- Average delivery time for on-time shipments: 1.84 days.
- Average delivery time for late shipments: 3.52 days.

Late shipments took approximately 1.68 days longer on average.

Carrier performance across warehouses

National Freight had consistently high late-delivery rates:

Warehouse	National Freight late rate
Birmingham	33.71%
Corby	39.29%
Leicester	42.53%

For comparison, RapidRoute recorded late-delivery rates of:

- Birmingham: 8.87%.
- Corby: 14.02%.
- Leicester: 21.90%.

Finding:

National Freight's weaker performance is not isolated to one warehouse. Its late-delivery rate is high across all three locations.

It also handles a relatively small share of shipments at each warehouse, approximately 19–23%, meaning its higher late-delivery rate is not simply explained by handling the greatest volume.

However, differences in delivery distances, destinations and service requirements have not yet been investigated.

Recommended follow-up:

Review National Freight's service performance and contractual expectations. Compare delivery areas, routes and service requirements before deciding whether carrier allocation should change.

3.5 Delivery attempts and late deliveries

Investigation: Compared delivery attempts against late-delivery rates, including a closer look at National Freight and Leicester.

Across all shipments:

Delivery attempts	Shipments	Average delivery time
1	822	2.16 days
2	299	2.33 days
3	261	2.31 days

The results show slightly longer average delivery times for shipments requiring additional attempts.

Further investigation found that National Freight's late-delivery rate increased with each attempt at Leicester:

- One attempt: 37.25%.
- Two attempts: 47.06%.
- Three attempts: 52.63%.

However, this pattern was not consistent across every carrier and warehouse.

Finding:

Repeat attempts are associated with higher late-delivery rates in some cases, but this does not prove that additional attempts cause delays. More difficult deliveries may be more likely to require repeat attempts in the first place.

Recommended follow-up:

Investigate failed first-delivery reasons, customer availability, address accuracy and delivery scheduling. Monitor first-attempt delivery success as a separate operational measure.

4. Business recommendations

The findings suggest five areas where Northstar Logistics could focus its attention.

Priority	Area	Recommended action
High	National Freight	Review carrier performance, service agreements and delivery routes.
High	Leicester warehouse	Investigate dispatch processes and carrier handover performance.
High	Inventory continuity	Review unexplained stock movements and adjustment records.

Medium	Customer cancellations	Investigate cancellation reasons for customers with recurring cancellations.
Medium	Delivery attempts	Monitor first-attempt delivery success and investigate failed deliveries.

These are recommendations for further investigation rather than confirmed solutions. The available data identifies patterns but does not establish their underlying causes.

5. Data limitations

The analysis has several limitations that should be considered when interpreting the results.

- **Cancellation reasons:** The dataset records whether an order was cancelled but does not explain why.
- **Delivery and cancellations:** Cancelled orders did not have associated shipment records. Therefore, the analysis cannot establish whether delivery delays caused cancellations.
- **Inventory movements:** The available inventory data does not fully explain every change between daily opening and closing balances.
- **Carrier allocation:** Delivery distance, route difficulty and destination characteristics have not been considered.
- **Small sample sizes:** Some customer and delivery-attempt groups contain relatively few records, making percentage comparisons less reliable.
- **Service-level definitions:** A delivery within one day should not automatically be treated as meeting a contractual Next Day SLA without confirming the relevant service agreement.

These limitations highlight areas where additional operational data would improve future analysis.

6. Technical skills demonstrated

This project used SQL to investigate realistic warehouse and logistics problems.

Techniques used included:

- `SELECT`, `WHERE`, `ORDER BY` and `GROUP BY`.
- Aggregate functions including `COUNT()`, `SUM()` and `AVG()`.
- Conditional aggregation using `CASE WHEN`.
- Joining multiple tables using `INNER JOIN` and `LEFT JOIN`.
- Date calculations using SQLite's `julianday()` and `strftime()`.
- Window functions, including `LAG()`.
- Percentage calculations and operational KPI comparisons.

- Data-quality checks and inventory reconciliation.
- Translating query results into business findings and recommendations.

7. Conclusion

The Northstar Logistics project demonstrates how SQL can be used to investigate operational performance across inventory, orders, warehouses and carriers.

The most significant findings were the unexplained inventory continuity issue, Leicester's higher late-delivery rate and National Freight's consistently weaker delivery performance.

The project also demonstrates the importance of looking beyond headline figures. Comparing warehouse performance, carrier allocation, delivery attempts and customer cancellation patterns provided a more detailed understanding of where potential operational issues may exist.

The next step would be to combine these findings into a simple operational dashboard, allowing a logistics or WMS support team to monitor stock accuracy, cancellations and delivery performance over time.